

2. First Order ODE

Bernoulli's eqn

$$\frac{dy}{dt} + p(t)y = g(t)y^n$$

Sol'n

$$\text{solve } v' + (1-n)pV = (1-n)g$$

where:
 $v = y^{1-n}$

*Can use integrating method

Seperable ODE

$$\frac{dy}{dx} = f(y)g(x)$$

Sol'n:

$$\frac{dy}{f(y)} = g(x)dx$$

$$\int \frac{1}{f(y)} dy = \int g(x) dx$$

Integrating Factor

$$\frac{dy}{dt} + p(t)y = g(t)$$

Sol'n:

$$y = \frac{1}{N}(c + \int Ng)$$

where:

$$N = e^{\int p dt}$$

Homogenous ODE

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right)$$

$$\text{ie: } \frac{dy}{dx} = \frac{y^2}{x^2 + y^2}$$

Sol'n:

replace y with xV

$$y = xV$$

Now $\frac{dV}{dx} = \frac{x^2 V^2}{x^2 + x^2 V^2}$

$$\frac{dV}{dx} = \frac{V^2}{1 + V^2}$$

then:

$$xV' = \frac{V^2}{1+V^2} - V \text{ SEPERABLE}$$

solve this

$$\frac{(1+V^2)dV}{(V^2-V-V^3)} = \frac{1}{x} dx$$

Possible ODE's (1st order)

① $\frac{dy}{dt} + pY = g$ | Integrating factor

② $\frac{dy}{dt} + pY = gY^n$ | Bernoulli's

③ $\frac{dy}{dx} = f(x)g(y)$ | Seperable

④ $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$ | Homogenous

⑤ $\frac{dy}{dt} = y^2$ | Non-Linear

Model 1 Linear

$$y' = ry$$



Model 2 Logistic

$$y' = ry\left(1 - \frac{y}{k}\right)$$



Model 3 Threshold

$$y' = -ry\left(1 - \frac{y}{T}\right)$$



Interval of Existence

- get restrictions on variable.

ie: $x \neq \pm 1$, then start at IC

ie: $IC = 0$, then interval is

$$-1 < x < 1$$

if $IC = 5$, then $1 < x < \infty$

Applications

Bank:

$$\frac{ds}{dt} - rs = k$$

where:

$$p = -r \quad N = e^{\int p dt} = e^{-rt}$$

Sol'n:

$$s = Ce^{rt} - \frac{k}{r}$$

$$= \left(S_0 + \frac{k}{r}\right)e^{rt} - \frac{k}{r}$$

Non-Linear ODE

$$\text{ex: } y' = \frac{t^2}{y^2-1}; y(0) = 1$$

What determines interval:

- ① IC
- ② Equation
- ③ Solution

Critical Points

$$\frac{dy}{dx} = f(y)$$

Crit points where $\rightarrow f(y) = 0$

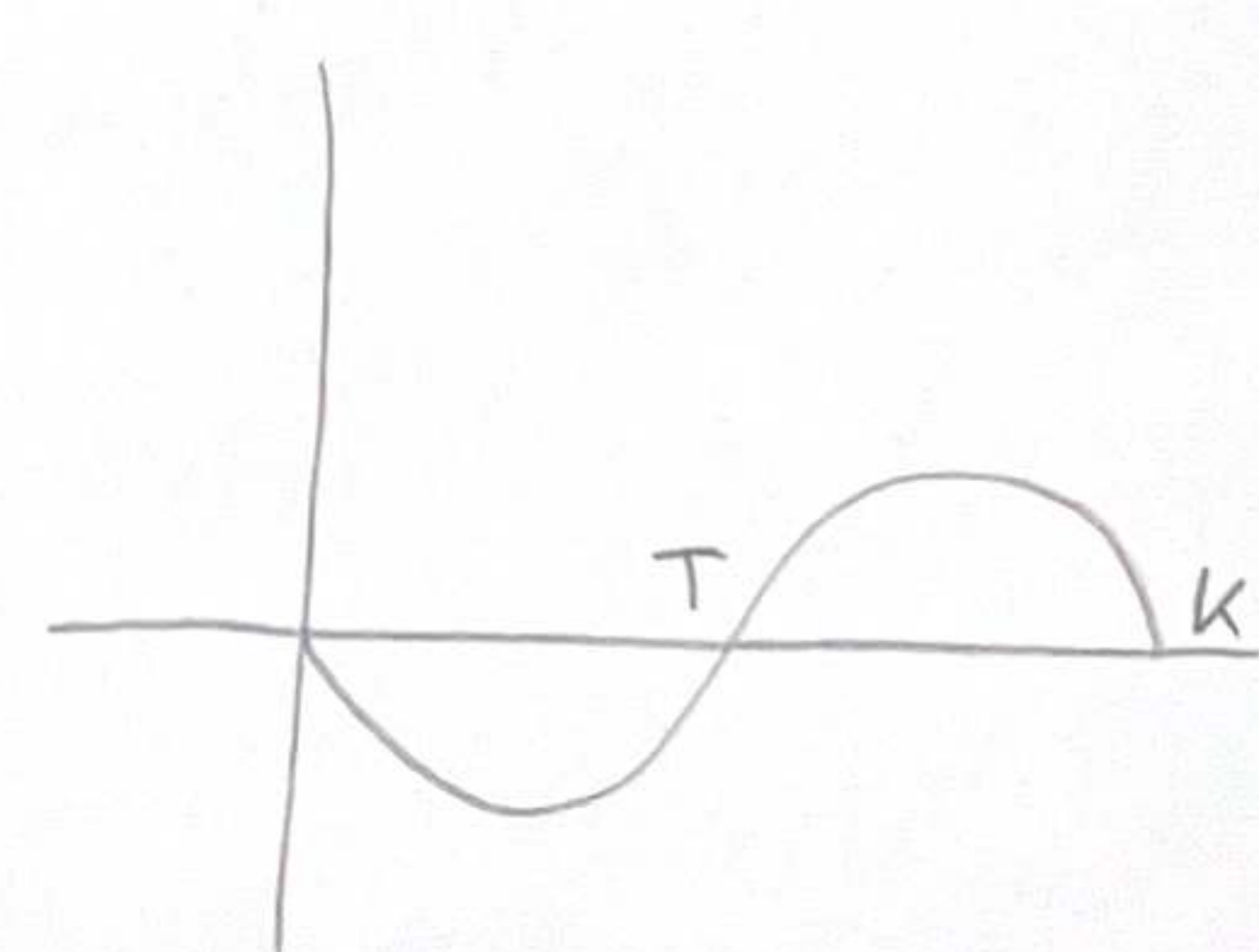
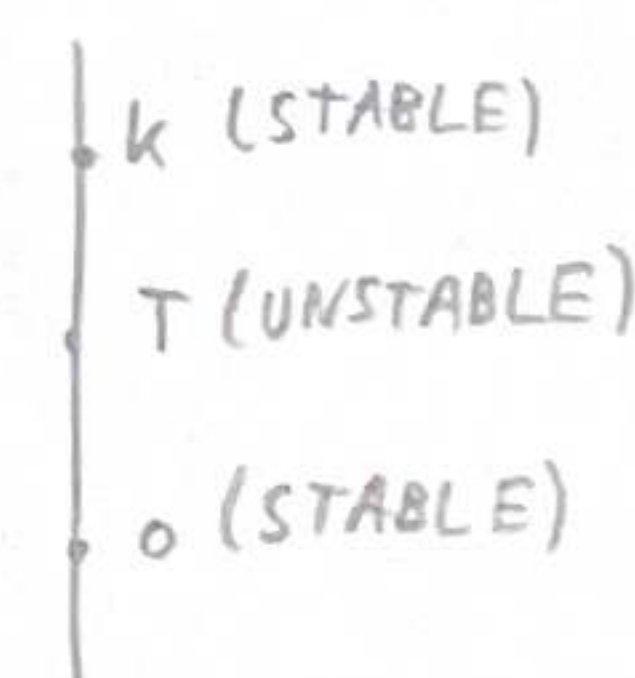
Case 1 $f'(y_0) < 0$ stable

Case 2 $f'(y_0) > 0$ unstable

Autonomous ODE and Population

Model 4 Logistic with Threshold

$$y' = -ry\left(1 - \frac{y}{T}\right)\left(1 - \frac{y}{k}\right) \quad 0 < T < k$$



3 Second Order ODE

Standard Form

$$y'' + p(t)y' + q(t)y = g(t)$$

Case 1 Constant Coefficients

$$ay'' + by' + cy = 0$$

$$ar^2 + br + c = 0$$

$$y_1 = C_1 e^{r_1 t} \quad y = a e^{r_1 t} + b e^{r_2 t}$$

$$y_2 = C_2 e^{r_2 t}$$

Case 3 Repeated Roots

$$ay'' + by' + cy = 0$$

$$ar^2 + br + c = 0$$

$$*r_1 = r_2 \quad y = C_1 e^{r_1 t} + C_2 t e^{r_1 t}$$

$$y_1 = C_1 e^{r_1 t}$$

$$y_2 = C_2 t e^{r_1 t}$$

Case 5 Euler Type

$$t^2 y'' + 3t y' + y = 0$$

$$at^2 y'' + bty' + cy = 0$$

$$ar(r-1) + br + c = 0$$

$$y_1 = C_1 e^{r_1 t} \quad y = C_1 e^{r_1 t} + C_2 e^{r_2 t}$$

$$y_2 = C_2 e^{r_2 t}$$

Method of undetermined coefficients

ex: $y'' - 4y' - 12y = 3e^{5t}$ *note:

$$r_1 = -2$$

$$r_2 = -6$$

guess $Y_p = Ae^{5t}$

$$Y_p' = 5Ae^{5t}$$

$$Y_p'' = 25Ae^{5t}$$

$$25Ae^{5t} - 4(5Ae^{5t}) - 12(5Ae^{5t}) = 3e^{5t}$$

$$-7Ae^{5t} = 3e^{5t}$$

$$-7A = 3$$

$$\therefore A = -\frac{3}{7}$$

$$Y_p = -\frac{3}{7}e^{5t}$$

$$y(t) = C_1 e^{-2t} + C_2 e^{-6t} - \frac{3}{7}e^{5t}$$

Wronskian

if $W \neq 0$,

y_1, y_2 are
fundamental
set of solutions

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$W = e^{-\int p dt}$$

Abel's formula:

$$W' + pW = 0$$

If one is true, all are true

$\{y_1, y_2\}$ F.S.S

$W \neq 0$

any sol'n can be written
as $ay_1 + by_2$

Case 2 Complex Roots

$$ay'' + by' + cy = 0$$

$$ar^2 + br + c = 0$$

$$y_1 = C_1 e^{(\lambda + iN)t}$$

$$y_2 = C_2 e^{(\lambda - iN)t}$$

$$y = C_1 e^{\lambda t} \cos(Nt) + C_2 e^{\lambda t} \sin(Nt)$$

Case 4 Reduction of order

"Non constant coefficients,
but we know y_1 "

$$W = e^{-\int p dt}$$

$$y = C_1 y_1 + C_2 y_2$$

$$y_2 = y_1 \int \frac{W}{y_1^2} dt$$

Case 6 Non-Homogenous

$$y'' + p(t)y' + q(t)y = g(t) \quad (1)$$

$$y'' + p(t)y' + q(t)y = 0 \quad (2)$$

y_1, y_2 are F.S.S to eqn (2)

Y_p is a special solution to eqn (1)

$$y(t) = C_1 y_1 + C_2 y_2 + Y_p$$

Method of Variation of Parameters

$$Y_h = C_1 y_1 + C_2 y_2$$

$$Y_p = u_1 y_1 + u_2 y_2$$

$$W = y_1 y_2' - y_2 y_1' \neq 0$$

To solve for u_1, u_2 :

$$u_1' y_1 + u_2' y_2 = 0 \quad (1)$$

$$u_1' y_1' + u_2' y_2' = g(t) \quad (2)$$

or directly:

$$Y_p = -y_1 \int \frac{y_2 g}{W} + y_2 \int \frac{y_1 g}{W}$$

where:

$$u_1 = -\int \frac{y_2 g}{W} \quad u_2 = \int \frac{y_1 g}{W}$$

$$y(t) = Y_h + Y_p$$

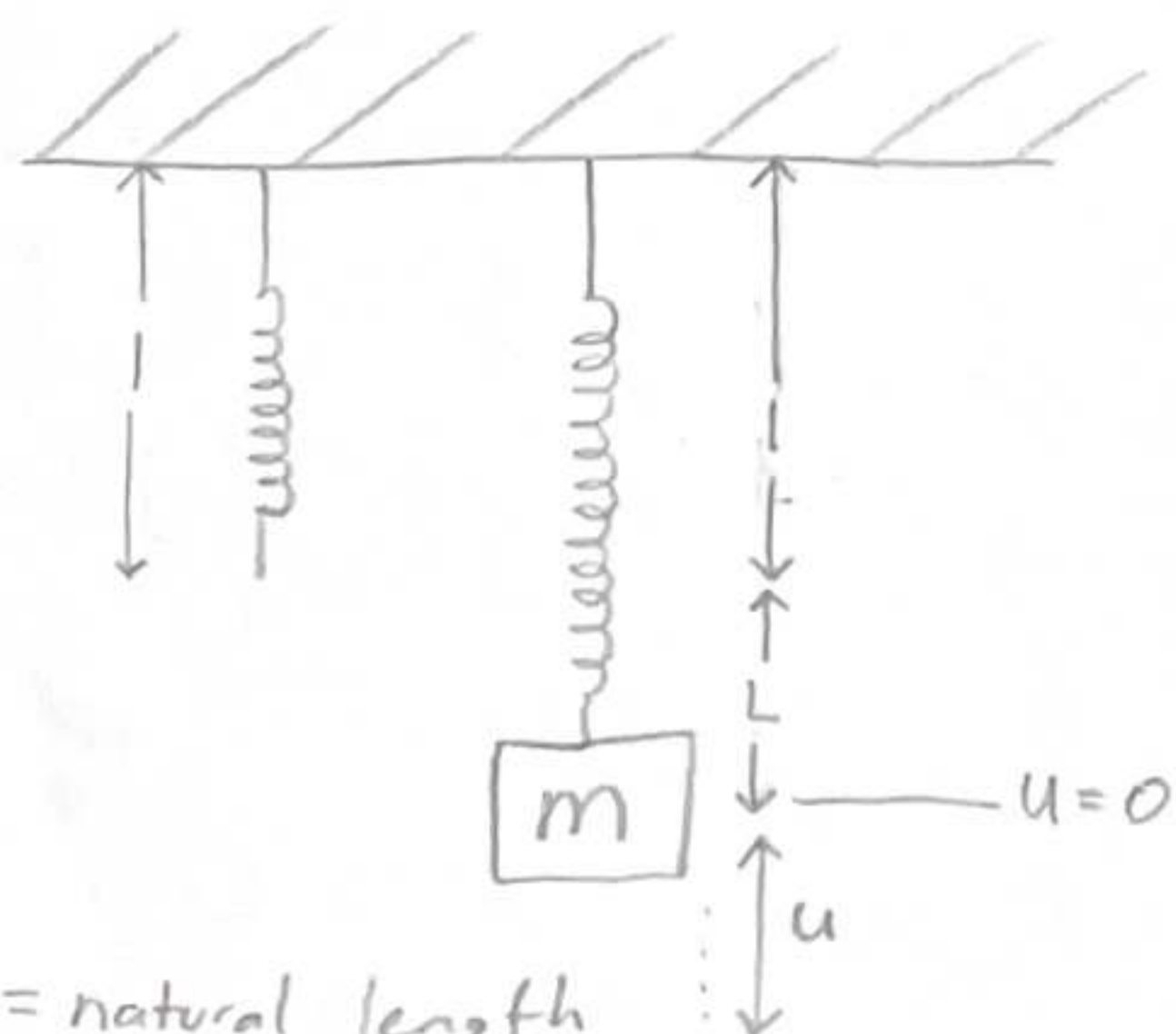
$$y(t) = C_1 y_1 + C_2 y_2 - y_1 \int \frac{y_2 g}{W} + y_2 \int \frac{y_1 g}{W}$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

Undetermined Coefficients Table

$g(t)$	$Y_p(t)$	Example
$e^{\alpha t}$	$t^s e^{\alpha t}$	$t^s A e^{\alpha t}$
$(\text{Polynomial})^n e^{\alpha t}$	$t^s (\text{Polynomial})^n e^{\alpha t}$	$t^s (At^2 + Bt + C) e^{\alpha t}$
$e^{\alpha t} \cos \beta t$ or $e^{\alpha t} \sin \beta t$	$t^s [A \cos \beta t + B \sin \beta t]$	$t^s (A \cos(4x) + B \sin(4x)) e^{6t}$
$(\text{Polynomial})^n e^{\alpha t} \cos \beta t$	$t^s [(\text{Polynomial})^n \cos \beta t + (\text{Polynomial})^n \sin \beta t] e^{\alpha t}$	$t^s [(At^2 + Bt + C) \cos(7t) + (Dt + E) \sin(7t)] e^{8t}$

Mechanical Vibrations



Known:

$$m = \frac{W}{g}$$

$$k = \frac{W}{L}$$

$$W = mg$$

$$* mg = kL$$

Form:

$$mu'' + \gamma u' + ku = F(t); u(0) = u_0, u'(0) = u_1$$

Case 1 (Free) $F(t) = 0$, (Undamped) $\gamma = 0$

$$mu'' + ku = 0$$

roots:

$$r = \pm i \sqrt{\frac{k}{m}}$$

$$r = \pm i \omega_0$$

where:

$$\omega_0 = \sqrt{\frac{k}{m}}$$

(natural frequency)

$$u = A \cos(\omega_0 t) + B \sin(\omega_0 t)$$

$$= R \cos(\omega_0 t - \gamma)$$

$$R = \sqrt{A^2 + B^2}$$

$$\gamma = \tan^{-1}\left(\frac{A}{B}\right)$$

$$\begin{cases} R \cos \gamma = A \\ R \sin \gamma = B \end{cases} \text{ SOLVE}$$

Case 2 (Free), $F(t) = 0$, (Damping) $\gamma > 0$

$$mu'' + \gamma u' + ku = 0$$

roots:

$$r_{1,2} = \frac{-\gamma \pm \sqrt{\gamma^2 - 4mk}}{2m}$$

3 scenarios:

① $\gamma^2 - 4mk > 0$ **OVERDAMPING**

$$u(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}$$

r_1, r_2 must be negative

So as $t \rightarrow \infty, u \rightarrow 0$

② $\gamma^2 - 4mk = 0$ **CRITICAL DAMPING**

$$u(t) = C_1 e^{\frac{-\gamma t}{2m}} + C_2 e^{\frac{-\gamma t}{2m}}$$

③ $\gamma^2 - 4mk < 0$ **UNDERDAMPING**

$$r_{1,2} = \lambda \pm Ni$$

$$u(t) = C_1 e^{\lambda t} \cos(Nt) + C_2 e^{\lambda t} \sin(Nt)$$

$$= R e^{\lambda t} \cos(Nt - \gamma)$$

Case 3 (Forced) $F(t) \neq 0$, (Undamped) $\gamma = 0$

$$mu'' + ku = F_0 \cos(\omega t)$$

2 Scenarios: By undetermined coefficients

① $\omega_0 \neq \omega$ $(-m\omega^2 A + kA) \cos \omega t + (-m\omega^2 B + kB) \sin \omega t = F_0 \cos \omega t$

Solving yields $A = \frac{F_0}{k - m\omega^2}$
 $B = 0$

So $u_h = C_1 \cos(\omega_0 t) + C_2 \sin(\omega_0 t)$

$u_p = \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t)$

$$u(t) = C_1 \cos(\omega_0 t) + C_2 \sin(\omega_0 t) + \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos(\omega t)$$

$$u(t) = R \cos(\omega_0 t - \gamma) + \frac{F_0}{m(\omega_0^2 - \omega^2)} \cos \omega t$$

② $\omega_0 = \omega$ (Resonance)

$$u(t) = C_1 \cos(\omega_0 t) + C_2 \sin(\omega_0 t) + \frac{F_0}{2m\omega_0} t \sin(\omega_0 t)$$

$$u(t) = R \cos(\omega_0 t - \gamma) + \frac{F_0}{2m\omega_0} t \sin(\omega_0 t)$$

l = natural length

L = displacement from l (Equilibrium position)

u = displacement from equilibrium length

Forces

$$F_g = mg$$

$$F_s = -k(L + u)$$

$$F_d = -\gamma \frac{du}{dt} \text{ (Friction)}$$

(Damping)

Case 4 (Forced) $F(t) \neq 0$, (Damped) $\gamma > 0$

$$mu'' + \gamma u' + ku = F(t)$$

u_h = Same as Case 2

u_p = determined by undetermined coefficients or variation of parameters

Systems of Equations

Case 1: Real Eigenvalues

$$X' = (A)X$$

$$\text{Sol'n } X = C_1(n^{(1)})e^{\lambda_1 t} + C_2(n^{(2)})e^{\lambda_2 t}$$

$$A\vec{n} = \lambda\vec{n}$$

where λ is an eigenvalue
 $\vec{n}$ is the eigenvector

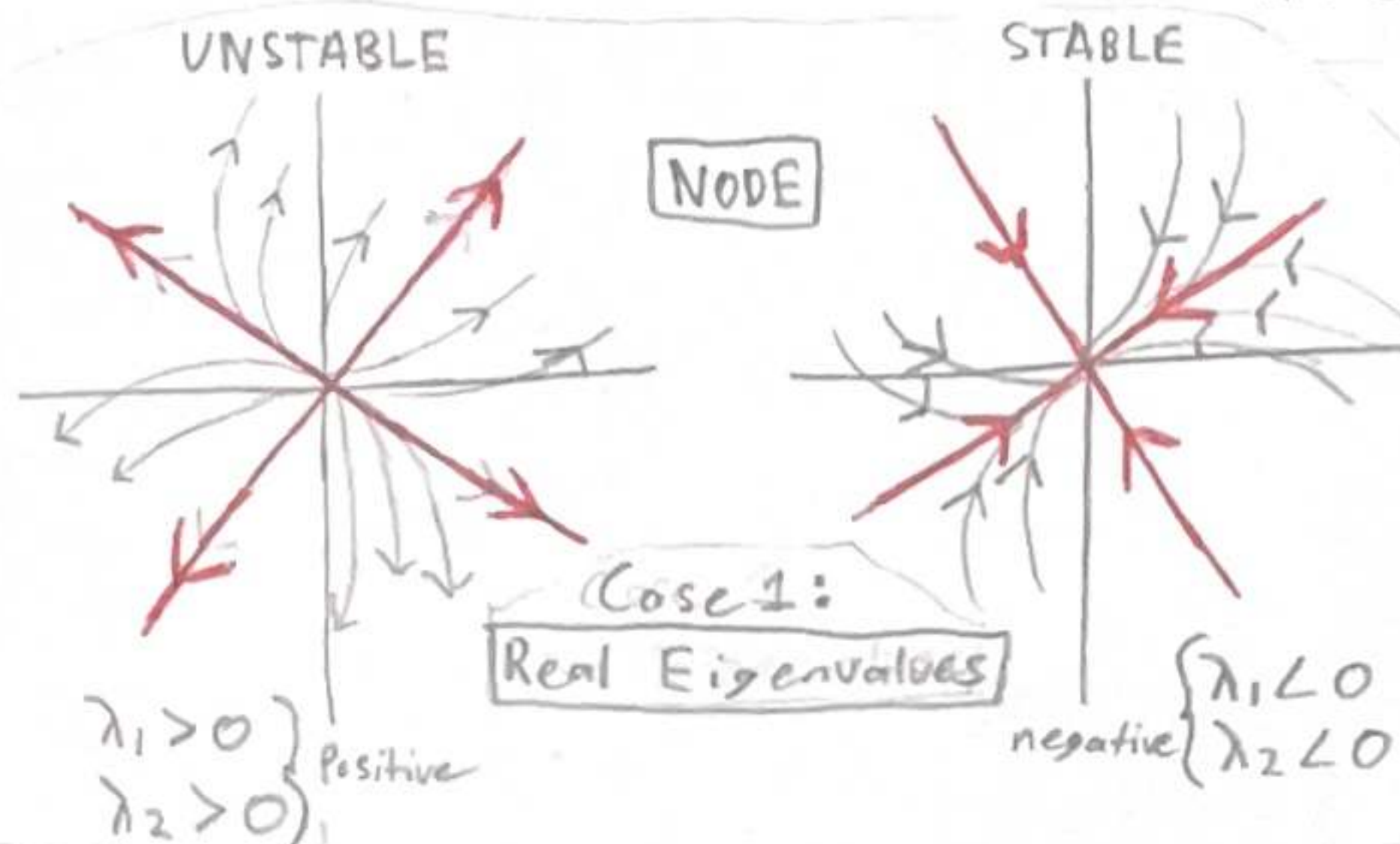
$$\det(A - \lambda I) = 0 \rightarrow \text{Characteristic Polynomial}$$

Solutions to System

$$X' = AX$$

$$\text{Sol'n } X = \vec{n}e^{\lambda t}$$

Phase Portraits



Method 1: Undetermined coefficients:

$$\text{① Solve homogeneous system}$$

$$\vec{X}_h(t) = C_1 \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{-t} + C_2 \begin{pmatrix} 2 \\ 3 \end{pmatrix} e^{4t}$$

Guesses

Case 3: Complex Eigenvalues

"When the real part is non-zero"

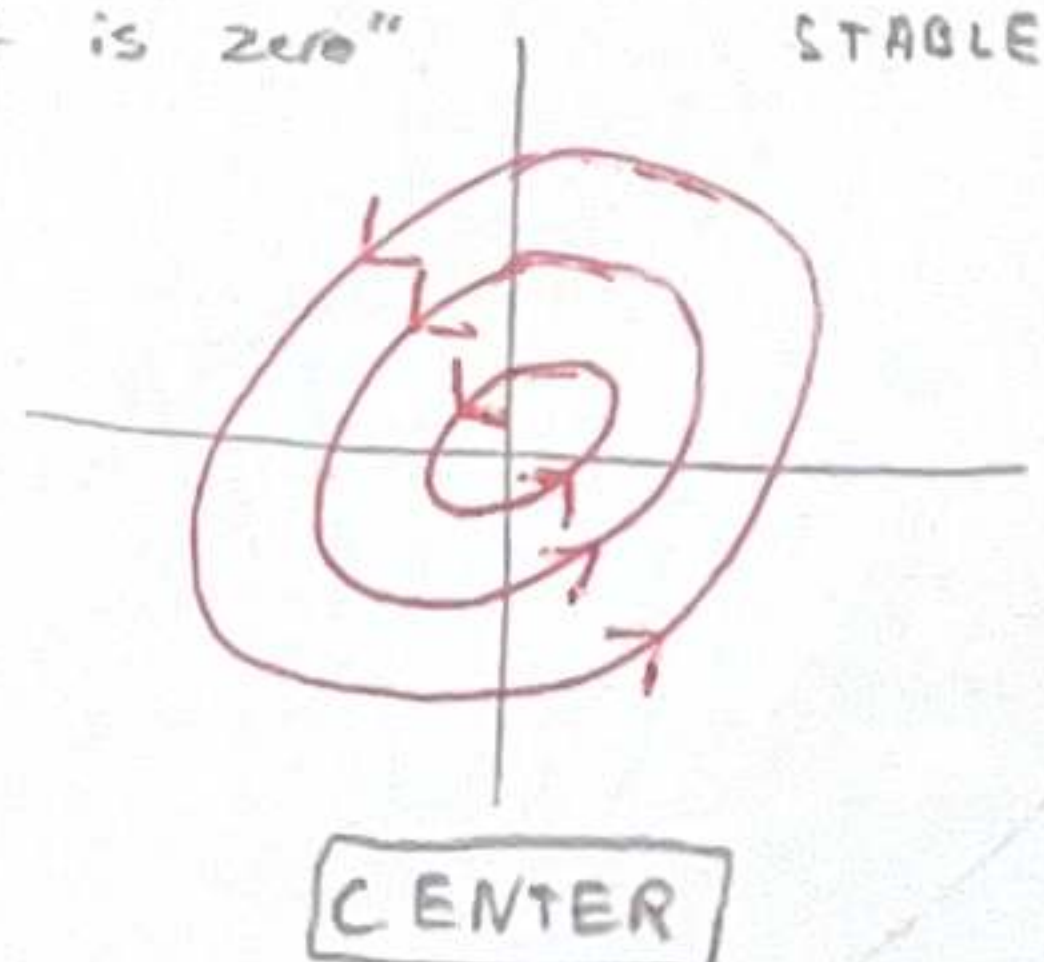


Method 2

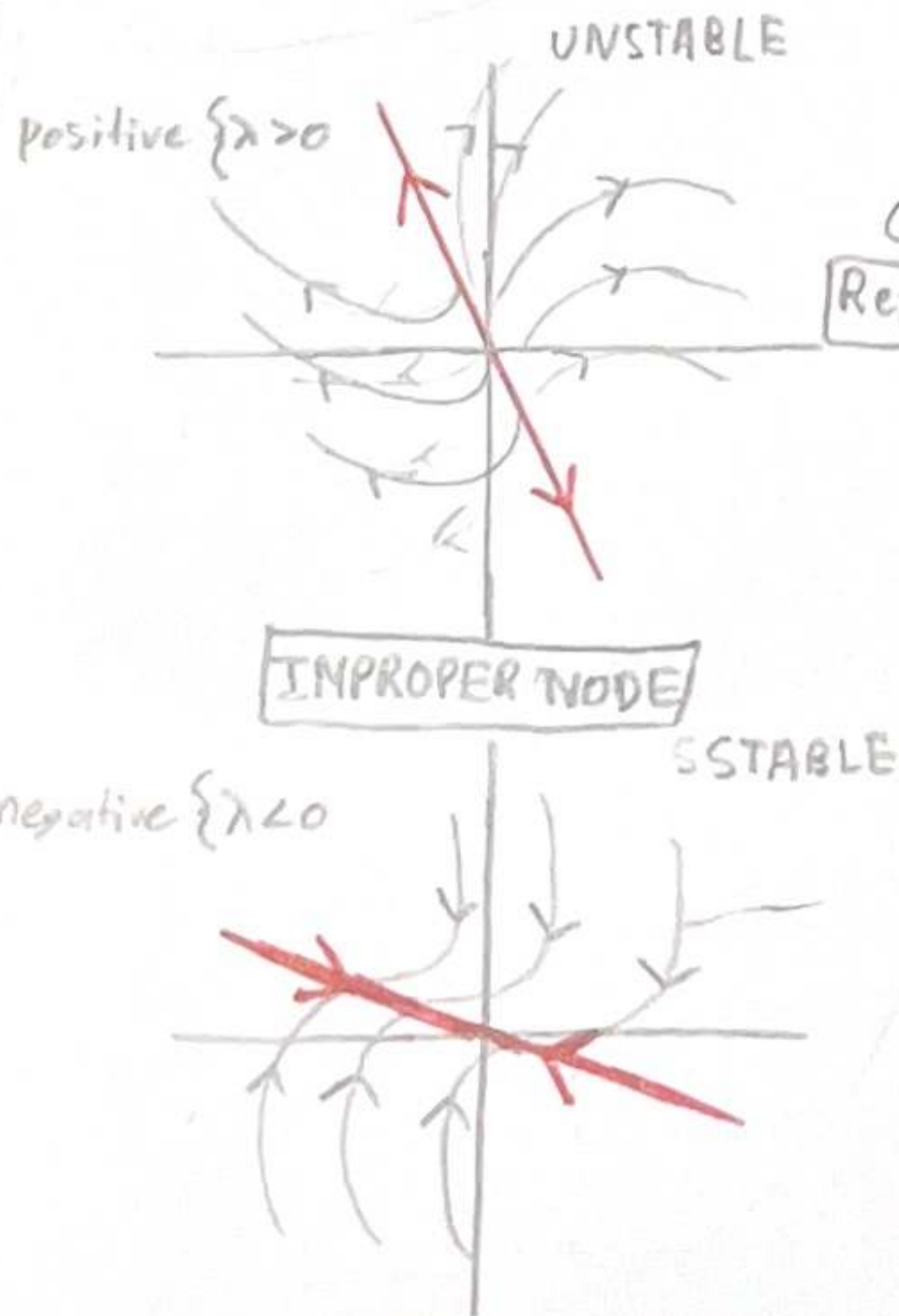
Variation of Parameter:

$$X_p = \Phi \left(\Phi^{-1} g dt \right)$$

"When the real part is zero"



Case 2: Repeated Eigenvalues



Boundary Value Problems

$$y'' + p(x)y' + q(x)y = g(x)$$

Homogenous when $g(x) = 0$

AND $y_0 = 0$ and $y_1 = 0$.

Otherwise $\rightarrow$ **Inhomogenous**

Periodic/orthogonal Functions

Periodic $\rightarrow f(x+T) = f(x)$ for all x

$\sin(wx)$ and $\cos(wx)$ are both

periodic functions with $T = \frac{2\pi}{w}$

even $\rightarrow f(-x) = f(x)$ ex: $x^2, \cos(x)$

odd $\rightarrow f(-x) = -f(x)$ ex: $x^3, \sin(x)$

Orthogonal

$f(x)$ and $g(x)$ are orthogonal on $a \leq x \leq b$

if
$$\int_a^b f(x)g(x)dx = 0$$

Types:

$$y(x_0) = y_0 \quad y(x_1) = y_1$$

$$y'(x_0) = y_0 \quad y'(x_1) = y_1$$

$$y'(x_0) = y_0 \quad y(x_1) = y_1$$

$$y(x_0) = y_0 \quad y'(x_1) = y_1$$

Ex: $y'' + 4y = 0; y(0) = -2, y(2\pi) = -2$

$$\therefore y(x) = C_1 \cos(2x) + C_2 \sin(2x)$$

$$\boxed{-2 = C_1}$$

$$\boxed{-2 = C_1}$$

$$\text{So... } y(x) = -2\cos(2x) + C_2 \sin(2x)$$

C_2 can be anything so we have ∞ many solutions

Ex: $y'' + 4y = 0; y(0) = -2, y(2\pi) = 3$

$$\therefore y(x) = C_1 \cos(2x) + C_2 \sin(2x)$$

$$\boxed{-2 = C_1}$$

$$\boxed{3 = C_1}$$

So... There are NO solutions

Trig Identities

$$\textcircled{1} \sin^2(x) + \cos^2(x) = 1$$

$$\textcircled{2} \sin(2x) = 2 \sin(x) \cos(x)$$

$$\textcircled{3} \cos(2x) = \cos^2(x) - \sin^2(x)$$

$$\textcircled{4} \sin(a) \cos(b) = \frac{1}{2} [\sin(a-b) + \sin(a+b)]$$

$$\textcircled{5} \sin(a) \sin(b) = \frac{1}{2} [\cos(a-b) - \cos(a+b)]$$

$$\textcircled{6} \cos(a) \cos(b) = \frac{1}{2} [\cos(a-b) + \cos(a+b)]$$

Extend ODD/EVEN

$f(x)$ is defined on $(0, L)$

$$f_{\text{even}}(x) = \begin{cases} f(-x), & -L < x < 0 \\ f(x), & 0 < x < L \end{cases}$$

$$f_{\text{odd}}(x) = \begin{cases} -f(-x), & -L < x < 0 \\ f(x), & 0 < x < L \end{cases}$$

Laplace Transformations

Linearity

Definition

$$L[af(t) + bg(t)] \\ = aL[f(t)] + bL[g(t)]$$

$f(t)$ is a piecewise continuous function

$$L[f(t)] = \int_0^{\infty} e^{-st} f(t) dt = F(s)$$

Table of Common Laplace Transformations:

$f(t) = L^{-1}[F(s)]$	$Y(s) = L[f(t)]$
1	$\frac{1}{s}$
e^{at}	$\frac{1}{s-a}$
$\sin(at)$	$\frac{a}{s^2+a^2}$
$\cos(at)$	$\frac{s}{s^2+a^2}$
$H(t-c)$ HEAVISIDE	$\frac{e^{-cs}}{s}$
$\delta(t-c)$ Dirac / Delta	e^{-cs}
$H(t-c)f(t-c)$	$e^{-cs} L[f(t)]$
$y'(t)$	$sY(s) - y(0)$
$y''(t)$	$s^2 Y(s) - sy(0) - y'(0)$
$t^n \quad n=1,2,3,\dots$	$\frac{n!}{s^{n+1}}$
t	$\frac{1}{s^2}$
$t^n f(t) \quad n=1,2,3,\dots$	$(-1)^n F^{(n)}(s)$

$$k u''_{xx} + f(x) = 0$$

Heat Equation

$$u_t = k u_{xx}$$

Dirichlet

$$\rightarrow u(x,t) = \sum_{n=1}^{\infty} a_n e^{-k \left(\frac{n\pi}{L}\right)^2 t} \sin\left(\frac{n\pi}{L} x\right)$$

$$u(0,t) = u(L,t) = 0$$

$$u(x,0) = f(x)$$

$$\text{where } a_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L} x\right) dx$$

$$u_t = k u_{xx}$$

Neumann

$$\rightarrow u(x,t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n e^{-k \left(\frac{n\pi}{L}\right)^2 t} \cos\left(\frac{n\pi}{L} x\right)$$

$$u_x(0,t) = u_x(L,t) = 0$$

$$u(x,0) = f(x)$$

$$a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi}{L} x\right) dx$$

$$u_t = k^2 u_{xx}$$

Periodic

$$u(x,t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} e^{-k \left(\frac{n\pi}{L}\right)^2 t} (a_n \cos\left(\frac{n\pi}{L} x\right) + b_n \sin\left(\frac{n\pi}{L} x\right))$$

$$u(-L,t) = u(L,t), u_x(-L,t) = u_x(L,t)$$

$$u(x,0) = f(x)$$

where a_n, b_n are from Formula sheet

Wave Equation

$$u_{tt} = c^2 u_{xx}$$

Dirichlet

$$u(x,t) = \sum_{n=1}^{\infty} (a_n \cos\left(\frac{n\pi}{L} ct\right) + b_n \sin\left(\frac{n\pi}{L} ct\right)) \sin\left(\frac{n\pi}{L} x\right)$$

$$u(0,t) = u(L,t) = 0$$

$$u(x,0) = f(x), u_t(x,0) = g(x)$$

$$u_{tt} = c^2 u_{xx}$$

Neumann

$$u(x,t) = \frac{a_0 + b_0 t}{2} + \sum_{n=1}^{\infty} (a_n \cos\left(\frac{n\pi}{L} ct\right) + b_n \sin\left(\frac{n\pi}{L} ct\right)) \cos\left(\frac{n\pi}{L} x\right)$$

$$u_x(0,t) = u_x(L,t) = 0$$

$$u(x,0) = f(x), u_t(x,0) = g(x)$$